

Oncology & Palliative Care

Friday November 28, 2025

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Overview (RC Objectives)

- Oncologic emergencies (1.4.12.3.1.)
- Screening and prevention (1.4.12.3.2.)
- Common solid tumour sites (1.4.12.3.4.)
 - **Breast**
 - **Lung**
 - **Colorectal**
 - **Prostate**
 - Cancers almost never tested but on objectives
 - Other GI (gastroesophageal, HCC, pancreatic)
 - Genitourinary (testes, bladder, renal cell)
 - Gynecologic (endometrial, cervical, ovarian)
- Systemic toxicities (1.4.12.3.3.)- Almost never tested except for immunotherapy toxicity
- Palliative care (1.4.13.7.)

Housekeeping

0-5% of Written Exam

- Screening, Diagnosis, Oncologic Emergencies, Palliative Care.
- Recognize certain treatment drugs and their S.E

Common Orals:

- Screening
- Diagnosing cancer
- Oncologic Emergencies can be a common oral scenario!

Remember: There are “trial” questions on RC each year- thus questions with no clear answers will generally not make it to subsequent exams



HIGH Yield for studying



Pearl- Exam Targeted Tips/ Trap Avoidance

NOT on Exam (NOE)



Context- (To help understanding)



Recent Changes



Controversial

Oncologic Emergencies

- Febrile Neutropenia, Hypercalcemia, TLS
- SVC, malignant bowel obstruction, Spinal Cord Compression, Brain Mets/Leptomeningeal Disease, Immune Toxicity



Q1 2026

A 62M with stage III colon cancer, on adjuvant 5FU/oxaliplatin, presents to the emergency department with T 38.2°C and absolute neutrophil count of 300/ μ L. He has a history of hypertension and mild chronic obstructive pulmonary disease (COPD). No known fungal infections. He reports mild fatigue but is alert, ambulatory, and tolerating oral intake. On exam he has a HR 88 bpm, BP 132/78 mmHg, RR 18, SpO₂ 96% on room air. No mucositis, no evidence of dehydration, and no localizing infection. He lives 10 minutes from the hospital, with a reliable caregiver.

- a. Admit for IV broad-spectrum antibiotics
- b. Outpatient management with oral broad-spectrum antibiotics and close follow-up
- c. Admit for observation without antibiotics unless he deteriorates, because fever may be viral.
- d. Discharge home without antibiotics, with return precautions



Febrile Neutropenia¹

Definition	Temperature Criteria	Absolute Neutrophil Count (ANC) Criteria
Both Criteria Need to be Met	- Single temp $\geq 38.3^{\circ}\text{C}$ OR $\geq 38.0^{\circ}\text{C}$ for $>1\text{hr}$	- ANC ≤ 1.0 - Moderate: 0.5-1.0 - Severe: ≤ 0.5 <i>Nadir of neutropenia occurs 7-14 days after chemo</i>
Work-up to find the infection (In addition to standard CBC, renal, liver function)	Standard: Blood culture x2 peripheral + any central lines (if present) - Urine culture - CXR Other: - Skin exam, Sputum (if present), Stool sample + C. diff (if diarrhea) - Additional imaging as indicated	
Management	Broad spectrum antibiotics IMMEDIATELY after cultures done Ex: Piptazo (Pseudomonas dose) to start, Add Vancomycin if still febrile after 72hrs	



Most source of Feb Neut is from patient's own gut
G-CSF used when neutropenic does NOT improve outcomes

MASCC Score

Clinical Characteristic	Score
Burden of illness: no or mild symptoms	5
Burden of illness: moderate symptoms	3
No hypotension (systolic BP ≥ 90 mmHg)	5
No chronic obstructive pulmonary disease (COPD)	4
Solid tumor or hematologic malignancy without prior fungal infection	4
No dehydration (requires no parenteral fluids)	3
Outpatient status at onset of fever	3
Age < 60 years	2

Low ≥ 21
High < 21

High Risk Patient → Admit for IV antibiotics



Outpatient management of Febrile Neutropenia in adults¹

Risk Stratification System	MASCC Score (COMMIT TO MEMORY)
Management	Administer empiric Antibiotics within 1 hour of triage • Ciprofloxacin + amoxicillin/clavulanate* or Clindamycin (if Pen allergic)
Bottom Line (Remember this is a safety exam): It's never wrong to ask Med Onc for help on oral exam. Consider Outpatient Management if all of the following- Picture a simple ED visit. • ID: Young Age- < 60 • Chief Complaint: Cancer that doesn't cause super low blood counts: Solid Tumor • PMH: No Comorbidities- No COPD/ Active Bronchitis • HPI: Fever Only- Minimal to no symptoms • Physical Exam: Unremarkable- No hypotension • Investigation: Unremarkable. Neutropenia will recover soon (Ex. chemo was 2 weeks ago) • Management: Easy. No fluids needed • Follow-up: Patient will be reliable in coming back if encounters problem.	



Neutrophils low beyond “nadir”- may be too much chemo was given/ patient factors



Hypercalcemia of Malignancy

Etiology	1. Lytic bone destruction • Breast Cancer, Myeloma 2. PTHrP production • Lung Cancer- Squamous cell - <u>NOT</u> Small Cell 3. 1,25-dihydroxyvit D/calcitriol production • Lymphoma
Management	• IV hydration (most effective short-terms) +/- Lasix • Calcitonin • Bisphosphonates • Zoledronic acid IV x 1 • Treat underlying malignancy



- Remember to reduce bisphosphonate dose for renal dysfunction
- Calcitonin IM/sc reserved for *severe* hypercalcemia – quick onset 4-6h but tachyphylaxis by 48h (use as a bridge to bisphosphonate effect)
- Bisphosphonates max effect 4-7 days post-infusion (therefore NOT short-term solution)
- Hypercalcemia can co-exist with active bone mets. Steroids are used to treat bone mets pain/ flare, not useful in lowering calcium.



Tumor Lysis Syndrome (TLS)

Etiology	Rapid Cellular Breakdown • Fast Cancer (Leukemia, Lymphoma), Large Cancer Burden – Spontaneous TLS • Sensitive to Treatment (Steroids, Chemo for Lymphoma) – Post-treatment TLS
Diagnosis	Requires ≥ 2 lab abnormalities within 7 days post-chemo 1. Hyper K⁺ – muscle weakness, ECG Δs, cardiac arrhythmias, sudden death 2. Hyper PO₄ – AKI, secondary hypocalcemia, N/V, lethargy, seizures 3. Hyperuricemia – AKI (renal tubule deposition), flank pain (renal stones), N/V, anorexia 4. Hypo Ca²⁺ – Tetany, paresthesias, mental status changes

Management of TLS

Treatment	Prophylaxis
• IV fluids (target U/O of 2cc/kg/hr) • Rasburicase ◦ Allopurinol if G6PD deficient. • Treatment of electrolyte abnormalities (Dialysis PRN)	• IV fluids <u>AND</u> • Allopurinol <u>OR</u> • Rasburicase (if NOT G6PD deficient) <u>OR</u> • Febuxostat



G6PD Deficiency – X linked recessive. If MCQ says a woman, no FamHx, assume no G6PD defic!



Malignant Bowel Obstruction (MBO)

Etiology	1. Intraluminal Block (Colon CA) 2. Peritoneal disease (Ovarian, Gastric metastases) 3. Extrinsic Compression (Intra-abdominal lymphoma/ Sarcoma)	
Diagnosis	CT abdomen	
Management	Interventional	• Consult Gen Surg (even if palliative) • stent, diverting ileostomy, venting G-tube
	Non-Pharm	• IV fluids to supplement losses • NG decompress
	Pharmacologic	• Octreotide ↓ gastric secretions, ↓ motility, ↓ splanchnic blood flow • ± Corticosteroids ↓ peritumor edema, ↑ mobility • Metoclopramide*



*For RC: Can use **if partial MBO**. **NOT** for use in **complete MBO**



SVC Syndrome

Etiology	Malignancy (up to 90% of cases)- Non-small cell lung cancer (~50% of cases) Thrombosis/ Fibrosis
Symptoms	Facial and arm edema, distended neck and chest veins, Dyspnea, cough, facial plethora, hoarseness, stridor, neurologic (confusion/change in LOC)
Diagnosis	CT chest, head and neck. Venography if unstable

Management of Tumor associated SVC Syndrome

Step 1	Be Safe. Call for help: Thoracics and Radiation Oncology	
Step 2	Life-threatening/ Grade 4 symptoms*	• Urgent Stent (Fastest) • Steroids if not able to stent or steroid-responsive histology (ex. Lymphoma) • Radiation to Temporize
	Non-life threatening	• <u>Obtain tissue diagnosis</u> to guide treatment. • Above options + Tumor-guided treatment (Radiation/ Systemic Therapy)

***Grade 4:** Life-threatening: confusion/obtunded, stridor, hemodynamic compromise (syncope without precipitant), hypotension



Spinal Cord Compression

Etiology	1. Vertebral fracture due to metastasis (Breast, Lung, Prostate, Multiple myeloma) 2. Tumor pushing into thecal sac (Metastasis or primary tumor ex. Lung)
Symptoms	• EARLY CLINICAL RECOGNITION IS KEY • Back pain (often 1st sign, occurs in 95% cases) • Leg weakness, sensory loss • Urinary retention, Bowel incontinence • If spinal cord compression = UMN findings* • If cauda equina syndrome = LMN findings, Saddle anesthesia*
Diagnosis	MRI <u>Whole Spine</u>

Management of Cord Compression

Step 1	Be Safe. Call for help: Spine Surgery <u>AND</u> Radiation Oncology (Rad Onc first if you have to pick one)
Step 2	• Steroids – Dexamethasone IV (Reduce Swelling/ Inflammation)** • Pain control



Have low threshold for scanning, even if neurological exam not definitive





Brain Mass/ Leptomeningeal Disease

Etiology	- Intra-parenchymal lesions causing vasogenic edema, or meningeal disease block CSF flow - Most common implicated malignancies – Breast, Lung, Melanoma metastases, Glioblastomas.
Symptoms	EARLY CLINICAL RECOGNITION IS KEY - Vertigo, Ataxia, Headache, Vision changes, Neurologic deficits
Diagnosis	Brain Mass: URGENT MRI Brain Leptomeningeal disease: URGENT MRI Brain + Full Spine. Consider LP depending on clinical status

Management of Brain Malignancy/ Leptomeningeal disease

Step 1	Be Safe. Consult Neurosurgical service <u>AND</u> Rad Onc*
Step 2	Steroids– Dexamethasone IV (Reduce Swelling/ Inflammation)** Further management (including mannitol or acetazolamide)- discuss with specialists



Q: UNRELENTING HEADACHE x days-weeks in a patient with active/ recent cancer within 5 years (even if hx of prior similar headaches)

A: MRI brain ideally, CT head w contrast if MRI not available, **contrast preferred w CT to r/o CVT**



*Both services can treat brain mets definitively. **(NOE)**

***Leptomeningeal disease**- Limited role for any service except treat underlying disease **(NOE)**



Immune Therapy Related Adverse Events (ITRAEs)

Definition/ Pathophys	- Immunotherapy (typically given IV) harnesses/modifies the patient's immune system to recognize and attack disease, most often cancer. It is <i>NOT</i> conventional chemotherapy. - ITRAEs stem <u>from immune overactivation</u>, leading to inflammatory or autoimmune toxicities affecting ANY gland/organ which can be life threatening Most commonly prescribed are: Pembrolizumab, nivolumab, durvalumab
Symptoms	- Vague, nonspecific symptoms such as fatigue, rash, diarrhea, cough, or dyspnea. - Can mimic infections or autoimmune disorders.
Diagnosis	Depending on the organ involved: elevated troponin (myocarditis), CK (myositis), transaminases (hepatitis), high creatinine (nephritis), thyroid dysfunction (TSH, free T4), lipase (pancreatitis), adrenal insufficiency (low Na, cortisol), or hyperglycemia (new-onset diabetes) or new DKA, etc...
Management	- Recognize, grade severity (Source: CCO ITRAE Toolkit) - Rule out infection, other etiologies - Mild cases= monitor and admit - Moderate to Severe= hold immune therapy, start corticosteroids +/- escalation to other immune suppressants if refractory - Consult appropriate subspecialty depending on affected organ (cardio, GI, endo etc.)

BONUS Q1 2026

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- a. **Admit for IV broad-spectrum antibiotics**
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- d. Discharge home without antibiotics, with return if symptoms worsen

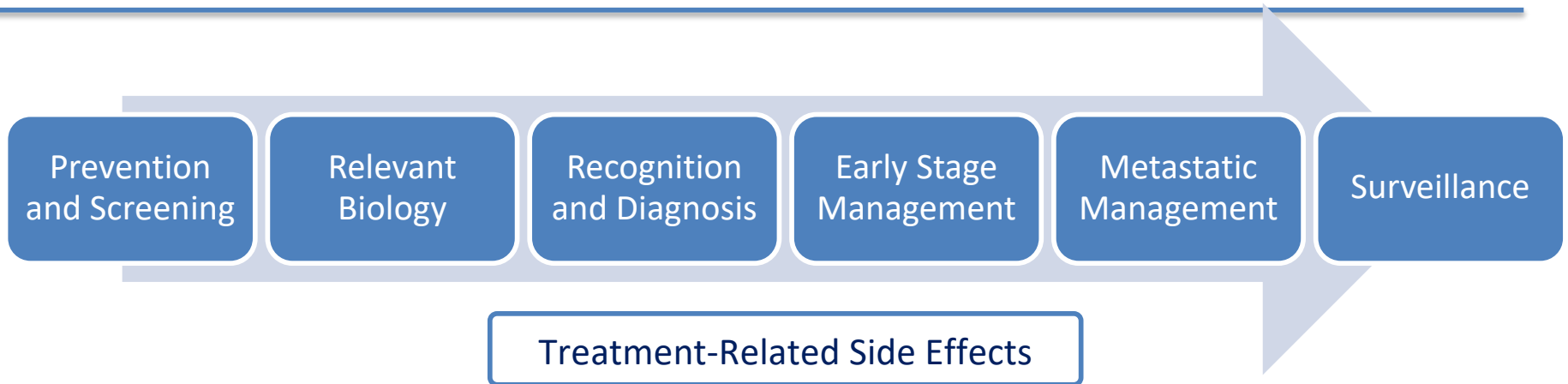
MASCC score breakdown: Mild or no symptoms = 5 , No hypotension = 5 , No COPD → *but he has COPD, so 0 points* , Solid tumor, no prior fungal infection = 4, No dehydration = 3, Outpatient at onset = 3 , Age <60 = 0 (*he's 62*). Total = 20. Interpretation: <21 = high risk, so he requires inpatient IV antibiotics.
Correct answer: A

Screening and Prevention

- Screening for Breast, Lung, Colon, HCC, Cervical
- Cancer Prevention



Approach to the solid tumor oncology section



Disclaimer: Early-stage and metastatic treatment sequence/regimen: very complicated. **Will only cover selected topics.**



Don't **ROUTINELY** order **cancer-characterizing (ie. Staging)** work-up without confirmation of cancer
Examples: Tumor markers or full body CT/ PET without pathologic evidence of cancer (**Caveat: unknown primary**)



Screening (Asymptomatic) is DIFFERENT than Diagnosis.
Ex. If there's a problem (ex. Membranous nephropathy):
Diagnosis with appropriate work-up. **DO NOT** follow screening guidelines
(ie. Don't do fecal immunohistochemical test when you see rectal bleed, **NEED TO SCOPE**)

Screening for Breast Cancer²

	Average Risk (No Prior Breast CA, amongst other factors)	
Age Groups	40-49*	Benefits/Harms Discussion
	50-74	Screen
	75+*	No screening
Method of Screening	Mammogram q2-3 yrs * Caveat: Provide info to all. If wants to screen- can screen.	
Source	CTFPHC MAY 2024 Draft Recommendation ²	
Bottom Line: Women Age 50-74, screen with Mammogram q2-3 years		



*For Age 40-49: **NOE**

Canadian Cancer Society: Mammo q2years.

CTFPHC DRAFT (May 2024): *if after discussion patient wants mammo, q2-3y.*

Breast Screening age 40-49 Across Canada

[As CTFPHC guidelines are still in draft and the practice varies WIDELY across Canada this is not MCQ material]

CTFPHC: “based on the current evidence (trials, observational studies, modelling and a review on values and preferences), we suggest not to systematically screen with mammography.”

ALSO CTFPHC: “Because individual values and preferences may differ, those who want to be screened after being informed of the benefits and harms should be offered screening every 2 to 3 years (conditional recommendation, very low certainty)”

Ontario/British Columbia/Newfoundland/PEI/ New Brunswick/Nova Scotia:	Access to provincial screening, q2Y
Alberta	Women 40-44yo if starting then access to q1yr mammogram (otherwise 45-74yo q2yr)
Saskatchewan	45-74yo q2yrs (however, slowly phasing into eventually including as young as 40yo).
Manitoba, Quebec	50-74yo q2yr

Breast Cancer Screening for 2S/Transgender/nonbinary/Gender Diverse Patients

History of Chest Surgery?	No Chest reduction surgery or simple reduction mammoplasty: Screen per usual guidelines	Mastectomy bilateral Screening mammo not recommended, 'discuss w/ Healthcare provider' if concern re risk.
Chest (breast) Tissue present due to gender-affirming hormone therapy? (Transfeminine patients)	If taking Estrogen >5yrs – screen per Provincial Guidelines (e.g. in Ontario –eligible to self-refer for screening mammo q2y 40-74)	

Screening for Lung cancer³

Risk Criteria (Need to meet all 3)	1) Age	55 to 74
	2) Smoking History	≥ 30 pack-year*
	3) Smoking Status	• Current smoker OR • Quit within the past 15 years
Method of Screening	Annual low dose CT every year up to 3 consecutive years	
Source	CTFPHC 2016 ³	
Bottom Line: Anyone 55-74 AND 30+py, AND current/ quit within 15 years, screen with annual CT x 3		

CTFPHC Review Comment (Similar Choosing Wisely comment)

Screen if patient has reasonable life expectancy.
(ie. Don't screen in most stage 4 cancers, end stage organ disease, etc.)



Screening for Colorectal Cancer (CRC) ^{4,5}

	Average Risk ⁴		Increased Risk ⁵
Applicable Population	People age 50+ No previous CRC or polyps, no IBD, no FHx CRC		≥ 1 First-degree relative with colon cancer OR advanced adenoma
Age Groups	50-74	Screen	2 options: Whichever youngest Age 40 10 yrs before earliest age of relative's diagnosis
	75+	No Screening	
Method of Screening	2 Options - FIT q2 yrs - Flex sigmoidoscopy q10 yrs DO NOT USE colonoscopy		- Colonoscopy q5-10 yrs - FIT q1-2 yrs can be considered as 2nd-line/alternative
Source	CTFPHC 2016 ⁴		CAG 2018 ⁵
Bottom Line	Colon CA average risk- FIT q2 or Sigmoidoscopy q10 from 50 to 74		Colon CA high risk- Colonoscopy q5-10 either from either 40 or 10 years prior to earliest dx

Screening for Hepatocellular Carcinoma (HCC) ^{6,7}

Cirrhosis	
Setting	AGE of Screening
Childs Pugh A-B	Any Age
Childs Pugh C	Screen ONLY if transplant candidate ⁷

No Cirrhosis but is Hep B carrier (sAg+)	
Setting	AGE of Screening
Endemic Countries (male 40+, female 50+)	40+
FHx of HCC in 1 st degree relative	
All HIV co-infected patients	
African or North American blacks	20+
Hep D Co-infected	Any Age

Method of Screening	Ultrasound q6 months (CASL) ⁶ / Ultrasound + AFP q6 months (AASLD) ⁷
Source	CASL 2018 + AASLD 2023



CASL 2018 does not recommend AFP where US is available (Insufficient evidence in 2018) **(NOE)**

Screening for Cervical Cancer⁸

⌚ CTFPHC update is currently 'paused' – provincial guidelines vary (NOE)

Applicable Population	Anyone 25+ with a cervix	
Age Groups	25-69	Screen
	70+	No/ STOP Screening IF ALSO ≥ 3 negative tests in the last 10 yrs
Population Excluded	Never sexually active, Previous abnormal Pap tests, Immunocompromised (e.g. HIV, organ transplant, chemo, chronic corticosteroid use), Symptomatic of cervical cancer (e.g. abnormal vaginal bleeding), limited life expectancy	
Method of Screening	Cervical cytology q3 years (Pap Smears)	
Source	CTFPHC 2013 ⁸	
Bottom Line	Age 25-69 with cervix- Pap smear q3yrs until age 70 + 3 negative tests over last 1yrs	

🗨️ ACS (American Cancer Society Recommendation 2020): HPV test q5 years age 25-65. (NOE)

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Many provinces have switched to HPV testing-based cervical screening

Other provinces/territories: Pap based testing as per CTFPHC 2013
 Provider collected: cervical brushing / broom
 Self-collected: vaginal swab

Province / Territory	Who's eligible (start age)	Screening interval	Test type(s)	When to stop & special cases
British Columbia	Sexually active asymptomatic people with a cervix, aged 25-69	Every 3-5 years (depending on test/history)	Primary HPV testing, self- or provider-collected; cytology/co- test in some cases	Stop at 69 if adequate negative history; exceptions for immunocompromised
Ontario		Every 5 years if HPV-negative; q3y if immunocompromised	Primary HPV with reflex cytology; Pap during transition	Stop at 69 if adequately screened
Saskatchewan		Every 3 years	Pap test with HPV reflex testing	Stop at 69 if adequately screened
Northwest Territories		Every 3 years	Primary HPV testing	Details pending in 2025 guideline
Prince Edward Island		Every 5 years for average risk age 25-65	HPV PCR testing; self-sampling rollout	~65-69 depending on history; program specific
Québec	Pap test: start at 21; HPV: start at 25	Pap: q2-3y; HPV: q5y	Pap (still in use); HPV replacing Pap regionally	Stop at 65 if adequately screened; individualized >65 ²⁷

DO NOT SCREEN for These Cancers

Touch me and I'll turn you into PESTO	
Nothing up my rear end	<u>P</u> rostate
Nothing down my throat	<u>E</u> sophagus even if chronic GERD*
Don't touch me	<u>S</u> kin (Melanoma)
Don't fondle my family jewels	<u>T</u> estes and <u>O</u> varies

 CTFPHC releasing update 2026 (NOE)



*Exclusion:

- Pertains to 'average risk' patients – exceptions if Known familial cancer Syndrome (e.g. Peutz-Jeughers (routine endoscopy), BRCA (PSA prostate), many more
- Previously diagnosed Barrett's
- Alarm symptoms (dysphagia, odynophagia, weight loss, anemia, bleed, loss of appetite) (Technically not screening)

Q2 2026

According to the 2025 Canadian Task Force on Preventative Health care smoking cessation guidelines, which of the following best reflects the recommended use of behavioural interventions for tobacco cessation?

- a. Interactive computer-based or online programs are sufficient as a stand-alone intervention.
- b. Brief counselling by a health worker (as short as 30 seconds to 3 minutes) is effective and recommended for all tobacco users.
- c. Intensive behavioural support should be offered to all smokers regardless of interest or readiness.
- d. Behavioural interventions are no longer recommended if pharmacotherapy is available.

Cancer Prevention- ETOH (Jan 2023)⁹



Canada's Guidance on Alcohol and Health, 2023			
0 drinks per week Not drinking has benefits, such as better health, and better sleep.	No risk	0	
1 to 2 standard drinks per week You will likely avoid alcohol-related consequences for yourself and others.	Low risk	1 2	
3 to 6 standard drinks per week Your risk of developing several different types of cancer, including breast and colon cancer, increases.	Moderate risk	3 4 5 6	
7 or more standard drinks per week Your risk of heart disease or stroke increases.	Increasingly high risk	7 8 +	
Each additional standard drink Radically increases the risk of these alcohol-related consequences.			

WHO- World Health Organization

CCSA- Canadian Centre on Substance Abuse and Addictions

Cancer Prevention- Smoking (2025) ¹⁰



SSRIs are NOT effective in smoking cessation

What we know	Cigarettes or Non-Cigarette = Harmful
Pharmacologic	1. Bupropion → Helpful with depression. Avoid if history of seizure or eating disorder 2. Cytisine → Not yet widely available in Canada, emerging evidence 3. Nicotine Replacement Therapy (NRT) → Use patch + short acting, titrate to cravings 4. Varenicline → Highest quit rates. Safe in stable CVD, monitor mood, nightmares Combination therapy better than monotherapy Start before or on quit date Treat 12 weeks; extend if relapse risk high.
Non-Pharmacologic	• Behavioural interventions (Brief health-worker counselling 30sec to 3 min) • Interactive computer-based or online programs with additional behavioural support (needs to be WITH behavioural intervention, not alone) • More intensive behavioural support for those interested
E-Cigarettes for Smoking Cessation	• Suggest against e cigarettes except in certain circumstances: other interventions fail, refusing alternatives, or strongly prefers them. • If chosen, pts should be informed of the uncertainties related to e-cigarettes: lack of long-term safety data and that continued use of nicotine containing e-cigarettes maintains addiction.
Source	• CTFPHC New 2025 Guidelines on Smoking Cessation • CCS 2025 smoking cessation

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Key Knowledge around *Common* Solid Tumour Sites



Q3 2026

A 62-year-old man presents with fatigue, unintentional weight loss, and new-onset hypertension. Laboratory results show: Hemoglobin: 190 g/L, Hct: 58%, Serum calcium: 2.85 mmol/L (elevated) PTH: suppressed, Chest X-ray: unremarkable

He has a 40-pack-year smoking history and no prior hematologic disease. Abdominal imaging is ordered. Which of the following is the most likely underlying diagnosis explaining his presentation?

- A. Polycythemia vera
- B. Small cell lung cancer with ectopic ACTH production
- C. Renal cell carcinoma with paraneoplastic syndromes
- D. Multiple myeloma



Breast Cancer: Biology + Confirm Cancer

Relevant Biology ESTROGEN (Hormone-driven) vs HER2				Diagnostic Workup (NEED ALL)
	ER/ PR pos (+)	ER/ PR neg (-)	Imaging	1. Diagnostic bilateral mammogram 2. Ultrasound of breast + axilla*
HER2 +	Triple +	HER2 +	Biopsy Method	Core needle biopsy (US-guided)
HER2 -	Hormone +	Triple -	Markers	1. ER (estrogen hormone receptor) 2. PR (progesterone hormone receptor) 3. HER-2 (human epidermal growth factor receptor)



Mastitis not responding to antibiotics → BIOPSY to rule out Paget's/ Inflammatory Breast CA)
Post COVID vaccine- Mammogram/U/S prior or 4-6 weeks later to avoid reactive lymph nodes



In real world: U/S + Mammogram TOGETHER with Biopsy (**NOE**)

- Once Confirmed Localized Breast Cancer- **Move on to Surgery! (Before completion of staging)**

Breast Cancer: Localized Management + Staging

	Tumor (2 options)	Lymph Nodes (LN) (2 options)
Surgery	1. Mastectomy 2. Lumpectomy (Breast Conserving Surgery)*	1. Sentinel LN Biopsy (SLNB)** 2. Axillary LN Dissection (ALND)
Bottom Line	For early stage Breast CA: Tumor treatment (Mastectomy / Lumpectomy) + Lymph node treatment (SLNB/ ALND)	

Staging: Decision based on Surgical pathology results	
Stage 1	No Staging
Stage 2	Stage only if symptoms
Stage 3	STAGE - Bone scan + CT C/A/P



PET now funded in Ontario for early stage breast CA (Stage 2B and 3) and oligometastatic breast CA

Breast Cancer: Systemic Therapy

Hormone +	HER2 +	Triple negative
1. Endocrine therapy 2. CDK 4/6 Post-menopause: TAM or AI Pre-menopause : TAM*	1. Chemo AND 2. Anti-HER2~	1. Chemo 2. Immunotherapy

KEY DRUGS TO BE AWARE OF FOR EXAM

Endocrine	Chemo	CDK4/6	Anti-HER2	Immunotherapy	Anti-Resorptive Agents
• Tamoxifen • Aromatase Inhibitor	Anthracycline (-rubicin)	- Ciclibs (ex. Ribociclib)	Trastuzumab Pertuzumab	Other -mabs excluding the 2 Anti-HER2s (ex. Pembrolizumab)	Zoledronic Acid Denosumab



*Pre-menopausal adjuvant occasionally use AI + GnRH agonist

~Many new drugs now. One of which is Antibody Drug Conjugate = Antibody Tagged with Chemo

Breast Cancer: Surveillance¹¹

Method of Surveillance*	Lifestyle Modification post Breast Cancer (CMAJ 2017) ¹¹
• Annual Clinical (History and Physical) • Annual Mammogram	Live healthy- Just know cardiac guidelines**



** Examples:

- Prevent weight gain
- Exercise (150 mins/wk = **reduces breast cancer mortality**)
- No smoking. Minimal Alcohol
- Limit saturated fats and high-fat dairy products



*Recommend **AGAINST** surveillance blood work, bone scan, CT scans (Choosing Wisely ASCO)



Breast Cancer Treatment Side Effects

Endocrine Therapy		
Side Effect	Tamoxifen	Aromatase Inhibitor
Endometrial cancer	↑	None
Thrombosis	↑	None
Osteoporosis	↓ (relative)	↑
CV Risk	↓ (relative)	↑

Cardiomyopathy	
Anthracyclines (ex. Doxorubicin)	Anti-Her2 (ex. Trastuzumab)
Irreversible	Reversible
Late in course/ delayed	During treatment



Side Effects of TAM vs AI plagues RC takers every year. I would stick to the top 3. The CV risk option is always tricky. Pick TAM > AI in reducing CV risk.

Lung Cancer: Biology

Non-Small Cell

Squamous Cell

- Central lesions
- Strong link with **smoking**
- Unlikely EGFR/ALK mutated

Adenocarcinoma

- Peripheral lesions
- Has driver mutations (EGFR)

Rare(ish) tumor types

- Neuroendocrine
- Adenosquamous
- Sarcomatoid
- Large cell

Small Cell

- Central lesions
- Strong link with **smoking**
- Non-EGFR/ALK mutated
- Rapidly growing

Bottom Line



- The 4 S's of Lung Ca: **S**mokers get **S**quamous and the **S**peedy **S**mall cell
- Adeno: "**I Don't Know** why I got cancer..." – Lots of targetable mutation
- Neuroendocrine tumor of lung (RARE- know workup for carcinoid):
 - **PREOP do an Echo if suspicious**
 - Serum Chromogranin A
 - Urine 24-HR Urine 5-HIAA.
 - Gallium-PET scan (NOT Regular PET)



Lung Cancer: Paraneoplastic Syndromes

Cancer Type	Paraneoplastic Syndromes Highlights
SCLC	• SIADH • Lambert-Eaton Myasthenic Syndrome (LEMS) ○ <u>Anti-VGCC Ab*</u> : reduced presynaptic ACh release ○ Similarities to myasthenia gravis, but notably absent/decreased reflexes • Encephalomyelitis & sensory neuropathy ○ <u>Anti-Hu Ab*</u> cross reacts w/ both SCLC antigens and neuron-specific RNA-binding nuclear proteins • Cushing's syndrome ○ Ectopic ACTH production. NOT suppressed by dexamethasone supp test
Adeno NSCLC	• Hypertrophic osteoarthropathy ○ Clubbing + periosteal new bone formation of tubular bones ○ Symmetrical, painful arthropathy (ankles, knees, wrists, elbows)
Squamous NSCLC	• Hypercalcemia ○ PTHrP production (NOT measured in modern assays for PTH)

***Only order antibodies (serum + CSF) if suspecting the clinical syndrome.
NOT used as a screening test for paraneoplastic syndromes**



Lung Cancer: Workup

Diagnosis of Cancer	Staging once diagnosis is confirmed
1. CT guided biopsy (Interventional Rad) OR 1. EBUS (Endobronchial ultrasound) biopsy (Thoracics/ Resp)*	1. CT Chest Abdomen Pelvis** AND 2. MRI brain (preferred over CT) 3. ± Bone scan if symptomatic

Molecular Testing- Genetic alterations/ signs of immune system attacking cancer using the tumor sample

	Adenocarcinoma	Squamous	Small Cell
Biomarkers	1. EGFR (amongst many others) 2. PD-L1	PD-L1	None
Bottom line: Having molecular profile is crucial to treating non-small cell lung cancer			



"Typical" EGFR+ profile – Elderly, Female, Asian, Non-smoker, Adenocarcinoma



* Bx of metastases okay too but pleural effusion alone not enough (Not enough cells for more testing) **(NOE)**

PET/CT scan and mediastinal staging is done prior to curative treatment- Job of the oncologists **(NOE)

NSCLC: Staging

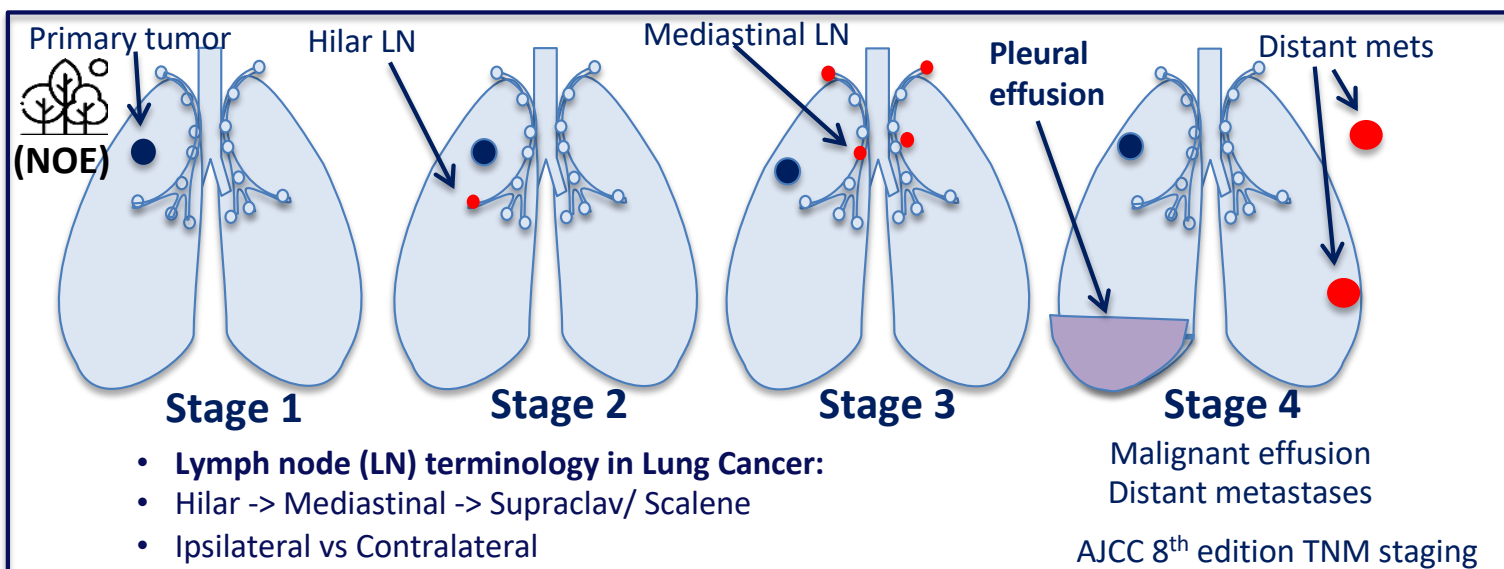
Malignant Pleural Effusion = Stage 4



Q: Symptomatic Pleural Effusion NYD for malignancy (Exudative doesn't count)?

A: Thoracics referral first (drain + diagnostic)

In real life may refer to both medical oncology and thoracics



Non-Small Cell Lung Cancer: Early Stage Management

	Stage 1	Stage 2 and 3 (Resectable)	Stage 3 (Unresectable)
Definitive Treatment	Fit for surgery: Surgery Unfit: Radiation (SBRT*) ¹²	Complicated**	Concurrent chemoradiation
Adjuvant Treatment	None		Immunotherapy x 1 yr

Bottom Line: For any lung **nodule** NYD **OR** diagnosed lung cancer with stage NYD (**PRESUMED EARLY STAGE**)...

Q: Next best service to consult?

A: Thoracics- Need to determine resectability (or Rad Onc if question says stage 1 and frail)

Q: One other service to consult aside from thoracics for stage 2 or 3?

A: Medical Oncology (See red context box below)

*SBRT = Stereotactic Body Radiation Treatment



****2025 Update: Definition of resectability is changing. Resectable stage 2 and 3 can be treated many ways**
Most commonly: neoadjuvant chemo + Immuno then surgery. EGFR+: surgery + adjuvant Osimertinib vs. neoadjuvant Osimertinib (NOE)

Non-Small Cell Lung Cancer: Metastatic Management

	EGFR mutation positive	Driver mutation negative
First Line Treatment	EGFR inhibitor*	Immunotherapy +/- Chemo ¹⁴
Bottom Line	Always refer to Medical Oncology first	



*Adenocarcinoma + EGFR? = **Osimertinib** (regardless of stage 2+ or late stage)
(As long as no mention of extreme frailty and refusing treatment)



Early Upfront involvement of palliative care for stage 4 NSCLC = survival benefit

2024 Update (NOE)



- Osimertinib + Chemo for EGFR. Other EGFR inhibitors may be better
- Double Immuno for lung CA
- Many targeted therapy for many mutations other than EGFR

Small Cell Lung Cancer: Workup & Management

	Limited Stage	Extensive Stage
Definition	Confined to 1 radiation field	1. Beyond 1 radiation field (i.e. mets) 2. Malignant Effusion
Treatment	Concurrent chemoradiation + consolidative immunotherapy +/- PCI*	Chemotherapy + Immunotherapy + Brain MRI Surveillance**
Bottom Line	Small Cell Lung Cancer can kill in short weeks to months. Refer to Medical Oncology and Radiation Oncology ASAP. Medical Oncology First	

 **2025 Update:** consolidation immunotherapy approved and funded for curative intent small cell lung cancer (NOE)

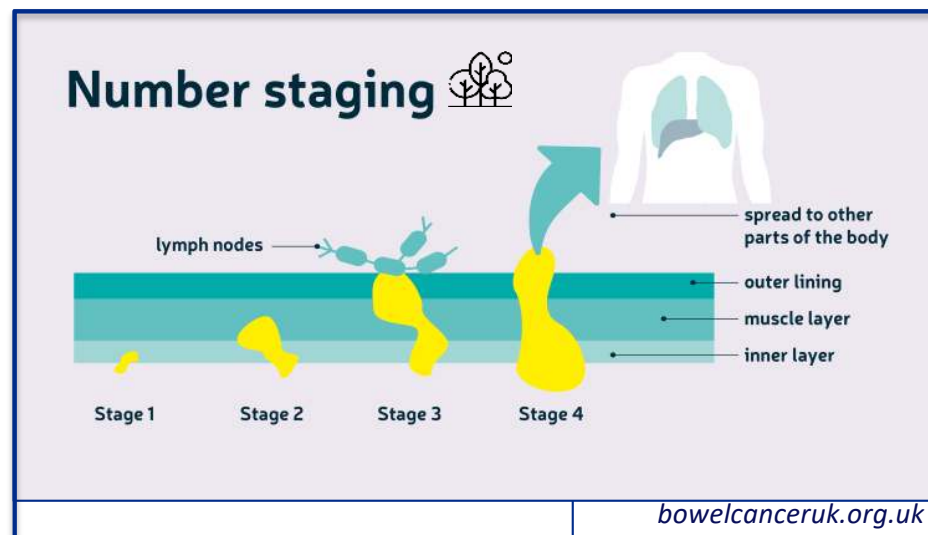
- Highly chemo/radio-sensitive, but frequently relapse
- **Surgery is generally NOT a part of SCLC treatment unless very early stage**
- *Prophylactic cranial irradiation (PCI) historically, however in the era of immunotherapy, less clear utility
- **More recent trials showed no OS benefit to PCI and more toxicity compared to MRI surveillance alone

 (NOE)

Colorectal Cancer: Workup



Diagnostic Method	Full Colonoscopy + Biopsy
Staging	CT Chest Abdomen and Pelvis
Markers	Carcinoembryonic antigen (CEA)



Colorectal Cancer: Management

	Stage 1 and 2 (No LNs)	Stage 3 (Has LNs)	Stage 4
<u>Treatment</u>	Surgery (Adjuvant Chemo for some stage 2)	Surgery + Adjuvant Chemo	Sytemic Therapy*
Bottom Line: - Upfront surgery for colon cancer - Stage 4 potentially curable (RARE in Solid Tumor Cancers)			



*Curable Stage 4

- Oligometastatic (stage 4): isolated liver or lung lesions, generally < 4 mets
 - Metastectomy + chemotherapy (May be curable)**



- Chemotherapy alone, Chemotherapy + VEGF antibody, Chemotherapy + EGFR antibody, Immunotherapy.
- Radiation limited role in all settings of colon cancer
- 2024 Updates: Double Immunotherapy for colon cancer
- 2024 Updates: Full liver transplant for liver-only metastases for colon cancer

(NOE)

Colorectal Cancer: Stage 1-3 Surveillance

🕒 CCO potential update 2025

Timeline	Year 1-3	Year 4-5	 CCO potential update
History and Physical	Q6 months	At discretion	
CEA	At discretion	At discretion	
CT CAP	CT CAP Year 1 and 3	At discretion	
Colonoscopy	• 1 year post-resection. • Subsequent colonoscopies based on findings of previous scope (not annually), if negative, every 5 years. ☞		
Bottom Line: Clinic visits q6 months for 3 years. CT year 1 and 3. C-scope at year 1 then go from there.			



Recurrence risk highest first 2-3 years. True Practice varies:

- Visits q6months x 3 years, annual at year 4-5.
 - CEA each visit.
 - CT CAP q6months x 2 years then annual to year 5.
- Follow-ups different for stage 1 vs stage 3

(NOE)

Gastroesophageal Cancer: Biology

	Squamous cell	Adenocarcinoma
Common location:	Upper-mid esophagus	Distal esophagus, stomach
Risk factors:	• <u>EtOH</u> • Caustic injury • Smoking	• Barrett's esophagus • GERD • Obesity • Smoking

 Leser-Trelat sign- **sudden** eruption of seborrheic keratosis. Think gastric or breast



Kilickap et al. NEJM 2007



Prostate Cancer: Biology + Workup

Biology	• Depends on Androgen • Metastasize to bone
Diagnosis	• Digital rectal examination • Prostate biopsy + <u>Gleason Score</u> (Calculated on biopsy specimen)
Markers	• PSA
Staging*	• Bone scan** • CT chest/abdomen/pelvis



Paraneoplastic coagulopathy. DIC most common in metastatic setting/ post-op



****Sclerotic Bone Lesion ONLY – Think Prostate Cancer Metastases**



***For High risk prostate CA only.**

Remember many prostate CA requires no treatment but needs to be watched

Prostate Cancer: Management

	Castrate Sensitive- Responds to lowering Androgen	Castrate Resistant
Early/ Localized	1. Active surveillance (PSA monitoring, intervene if progression)⌘ 2. Radical prostatectomy 3. Radiation (Brachytherapy or External Beam)	NoE
Metastatic	1. Androgen Deprivation Therapy (ADT) AND 2. Other Agents • CYP17 inhibitor (shuts down androgen production) • Anti-androgen pill +/- Chemotherapy	NoE



*ADT = Monthly injections to shut down androgen production.

2024 Update:



- Anti-androgen pill and CYP17 inhibitor moved into Early/Localized setting.
- PARP inhibitors (Used to be only for BRCA patients) may be used as front-line treatment for metastatic disease

BONUS

Read on own

Testicular Cancer

Biology	• Many types. Primarily would be tumor from germ cells (Germ cell tumors) • 2 types of Germ Cell Tumors: Seminoma vs Non-Seminoma
Diagnosis	• Scrotal ultrasound to find mass with radical orchiectomy
Markers (Need all 3)	1. β-hCG 2. AFP (NEVER elevated in seminoma. ONLY in Non-seminoma) 3. LDH
Staging	• CT chest/abdomen/pelvis
Management	• Localized: Surgery • Metastatic: Chemotherapy (Bleomycin)

1. **NEVER** needle biopsy testicular mass- risk of tumor seeding
2. If AFP elevated. non-seminoma will ALWAYS be present (worse survival, higher recurrence)
3. Patient with Pulmonary Fibrosis and any history of Testicular Cancer? A: Bleomycin toxicity
4. Testicular Cancer- Another rare solid tumor cancer that can be cured when metastasized.
5. Stage 1-3 only. There is no stage 4 (Metastatic disease is still stage 3)



BONUS

Read on own

Melanoma

Stage	Depth	Treatment
0	Epidermis	"melanoma in situ" – wide local excision
I	≤1mm thick	Wide local excision
II	>1mm thick	Wide local excision with sentinel LN biopsy, consider adjuvant for high risk (NoE)
III	Any thickness with regional nodal involvement or in-transit metastases	Excision + adjuvant therapy : Immunotherapy (e.g., nivolumab, pembrolizumab) or targeted therapy (BRAF/MEK inhibitors if mutation-positive).
IV	Distant metastases (brain, liver, lung...)	Immunotherapy (PD-1 inhibitors ± CTLA-4 inhibitors). Targeted therapy (BRAF V600E/K mutation: dabrafenib + trametinib). 



A symmetry
B order
C colour (uneven, brown, black, red... can be amelanotic also!)
D iameter - >6mm

BONUS
Read on own

Self Study: Renal Cell Carcinoma

RCC once called “The Internist’s Tumor” due to Paraneoplastic effects including Hypertension, Polycythemia, Thrombosis, *endovascular extension up IVC* ... also note seen w/ some genetic conditons ex Von Hippel Lindau

Localized renal mass

- Localized renal mass <1cm
 - **Active surveillance** (median growth rate = 2-3mm/yr)
- Localized renal mass 1 - 4cm
 - **Renal CT/MRI** to better characterize
 - Good surgical candidate → **partial nephrectomy**
 - Poor surgical candidate → **consider biopsy and ablation**
- Localized renal mass >4cm
 - **Nephrectomy**



NoE

Metastatic

- “Favorable risk”* → **Oral TKI** (Axitinib) + **Immunotherapy** (Pembrolizumab)
- “Intermediate/Poor risk”* → **Dual immunotherapy** (Ipilimumab/Nivolumab)



NoE

BONUS
Read on own

Self Study: Ovarian Cancer

Consider refer to genetics! Any ovarian cancer that is described as high-grade serous (this includes similar cancer in the fallopian tubes or the peritoneum) – 15-25% have germline BRCA1 or 2 mutation

- **Imaging**

- Abdominal + Transvaginal ultrasound **AND** CT chest, abdomen, pelvis

- **Tumour marker:** Serum CA-125

Typical regimen in adjuvant setting: platinum + taxane (Carboplatin + Paclitaxel)

- **Management**

- Localized

- **Surgery** (TAH + BSO + LN dissection + peritoneal washing)
- Do not biopsy adnexal mass = risk of seeding. Perform upfront surgery

- Stage III (visible peritoneal deposits +/- malignant ascites)

- **Cytoreduction** (debulk) → **Adjuvant chemo** (curative intent)

- Stage IV (mets beyond pelvis, malignant pleural effusion)

- **Palliative chemotherapy** (+/- debulking surgery) +/- **VEG-F inhibitor** (Bevacizumab) OR **PARP inhibitor** (Olaparib, if *BRCA* +)



Q3 2026

A 62-year-old man presents with fatigue, unintentional weight loss, and new-onset hypertension. Laboratory results show: Hemoglobin: 190 g/L, Hct: 58%, Serum calcium: 2.85 mmol/L (elevated) PTH: suppressed, Chest X-ray: unremarkable

He has a 40-pack-year smoking history and Abdominal imaging is ordered. Which of the following is the underlying diagnosis explaining his presentation?

- A. Polycythemia vera
- B. Small cell lung cancer with ectopic ACTH
- C. Renal cell carcinoma with paraneoplastic syndrome**
- D. Multiple myeloma

RCC = “The Internist’s Tumour”

RCC paraneoplastic syndromes: erythrocytosis (↑EPO), hypercalcemia (PTHrP), hypertension, Stauffer syndrome (non-metastatic hepatic dysfunction). Polycythemia vera (A) would not explain hypercalcemia with suppressed PTH. Small cell lung cancer (B) causes SIADH or ectopic ACTH, not erythrocytosis + hypercalcemia. Multiple myeloma (D) causes hypercalcemia but usually anemia, not polycythemia.

Systemic Therapy and their Side Effects

- Chemotherapy (Common, Life-Threatening)
- Immunotherapy



Q4 2026

A 45-year-old man with metastatic renal cell carcinoma presents with 2 days of polyuria, polydipsia, nausea, and abdominal pain. He has no prior history of diabetes. He is currently receiving pembrolizumab (last dose 3 weeks ago). On exam: HR 118 bpm, BP 102/66 mmHg, RR 26 with deep respirations, dry mucous membranes. Glucose: 25 mmol/L Na 130 mmol/L, K 4.9 mmol/L, HCO₃ 12 mmol/L, Anion gap: 22, Urinalysis pending.

What is the next best step?

- a) Steroid-induced hyperglycemia; start basal-bolus insulin and follow up outpatient
- b) Type 2 diabetes mellitus; start metformin and arrange endocrinology referral
- c) Diabetic ketoacidosis secondary to immune checkpoint inhibitor–induced autoimmune diabetes; admit for IV insulin, fluids, and urgent endocrinology involvement
- d) Hyperosmolar hyperglycemic state due to infection; give IV fluids only and monitor



The Terrible 2's: Heart and Lung

Organ	Heart		Lung
	Anthracyclines (-rubicins) Ex. Doxorubicin	Trastuzumab	Bleomycin
Reversibility	Irreversible	Reversible*	Irreversible
Onset	Delayed/ Cumulative with dose	Early onset/ During treatment	Delayed/ Cumulative with dose
Symptoms	Typical HF symptoms		Pneumonitis symptoms (Fatigue, cough, SOB)
Diagnosis	TTE		CT chest

* Real Life: Trastuzumab MAY be restarted after recovery of heart function

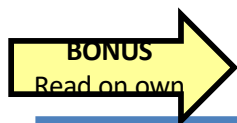
“Higher Yield Systemic Therapy Names to recognize”
(Scattered throughout the lecture)

Type of therapy	Drug Name (Drug Class)	Exam-Relevant Importance
Chemo	<i>Doxorubicin</i> (Anthracyclines)	• Cardiomyopathy
	<i>Bleomycin</i>	• Pneumonitis
Antibody (Anti-HER2)	<i>Trastuzumab</i>	• Cardiomyopathy
Anti-Hormone (Estrogen)	<i>Tamoxifen</i>	• Pre-menopausal uses • VTE. Endometrial CA
	(Aromatase Inhibitor)	• Post-menopausal uses • Osteoporosis
Androgen Deprivation Therapy	(ADT)	• Menopause for men symptoms
Anti-Androgen	<i>Abiraterone</i>	• Hyperaldosteronism
	<i>Enzalutamide / Apalutamide</i>	• Falls, seizures
Immunotherapy	<i>Pembrolizumab</i>	• Immunotherapy toxicities
	<i>Nivolumab</i>	
Targeted therapy (EGFR)	<i>Osimertinib</i>	• Very good drug for lung cancer with EGFR mutations. Select population

Common Chemo Side Effects: Nausea

Anti-Emetic	Chemo induced nausea (Starts 2-3 days post-chemo)		
	Mild	Moderate	Highly
NK₁-receptor antagonist (-pitant) Ex. Neuprepitant, aprepitant			✓
5-HT₃ antagonists (-setron) Ex. Ondansetron, granisetron	✓ OR	✓	✓
Steroid (dexamethasone)	✓	✓	✓
5-HT₂/D₂ antagonist (olanzapine)		✓	✓
Remember: Treat Nausea SOON! (Steroid, Ondans, Olanz, NK1)			

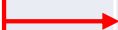
- **Anticipatory nausea (Vomiting before/during chemo)**
 - Refer to Psychiatry (behavioural therapy)
 - May use Benzodiazepines (**Lorazepam**)

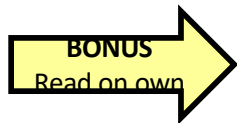


Chemotherapy Toxicities – REFERENCE ONLY, not comprehensive list of all treatments! See bonus slides for some other toxicity tips.

Drug	Toxicity	Common Use
Anthracyclines Doxorubicin (Adriamycin), Epirubicin	Irreversible cardiomyopathy Secondary leukemias	Breast CA (FEC-D, ddAC-T) Lymphoma (R-CHOP)
Antimetabolite 5-Fluorouracil, Capecitabine	Diarrhea, mucositis, hand-foot syndrome, coronary vasospasm	Colon CA (FOLFOX/FOLFIRI) Breast CA (FEC-D)
Topoisomerase inhibitors - Irinotecan - Etoposide	myelosuppression, diarrhea , nausea, vomiting, cholinergic reactions (Irinotecan) Etoposide: Hypersensitivity reactions	Colon CA (FOLFIRI) Testicular (BEP) Lung (small cell Cis + Etop)
Platinums - Cisplatin - Carboplatin - Oxaliplatin	All – Peripheral neuropathy Cisplatin-specific – Highly emetogenic, nephrotoxic, ototoxic, hypoMg/Ca/K Oxaliplatin-specific – Cold-induced neuropathy	Cisplatin – Lung CA, H & N Carboplatin – Ovarian CA Oxaliplatin – Colon CA
Taxanes - Paclitaxel - Docetaxel	Peripheral neuropathy , myalgias, hypersensitivity reaction (SOB, flushing, hypotension), fluid retention (docetaxel)	Breast CA (FEC-D, ddAC-T) Ovarian CA Prostate CA
Vinca alkaloids Vincristine, Vinorelbine	Peripheral neuropathy	Lung CA, Breast CA CHOP (“O”)
Cyclophosphamide	Emetogenic Hemorrhagic cystitis, bladder CA Secondary malignancies (MDS, AML, bladder Ca) Infertility	Breast CA (FEC-D, ddAC-T) CHOP (“C”)

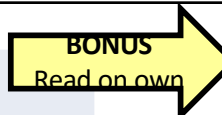
****GIM relevant****





Other Systemic Therapy Toxicities – REFERENCE ONLY

Drug	Toxicity	Common Use
Immunotherapy (not comprehensive) - Pembrolizumab, Nivolumab - Ipilimumab - Atezolizumab, Durvalumab	Autoimmune phenomena Most common are dermatitis, hypothyroidism. Rare: hypophysitis, pneumonitis, colitis	Melanoma, Lung Cancer Bladder Cancer, Renal Cell, Gastroesophageal
Anti-angiogenesis/VEGF inhibitors Bevacizumab (Avastin)	HTN, bleeding, bowel perforation , delayed wound healing, arterial thrombo-embolism (MI, CVA)	Metastatic colon CA Gynecologic CA
EGFR Tyrosine-kinase inhibitors Osimertinib	Acneiform rash Diarrhea ILD	Metastatic lung CA (NSCLC)
Trastuzumab (Herceptin)	Cardiomyopathy (reversible) Infusion-rxn (flu-like)	HER2+ breast CA
Rituximab	Infusion rxn (flushing, hypotension, tachycardia, allergic)	Lymphoma (R -CHOP)
Tyrosine-kinase inhibitors (TKIs) Sunitinib, Sorafenib	HTN , bleeding, hand-food skin reaction, diarrhea, mucositis, hypothyroidism (sunitinib)	RCC, HCC
PARP inhibitor Olaparib, Niraparib	Cytopenias GI symptoms	<i>BRCA</i> + Ovarian cancer, Pancreatic cancer



Toxicities of Radiation

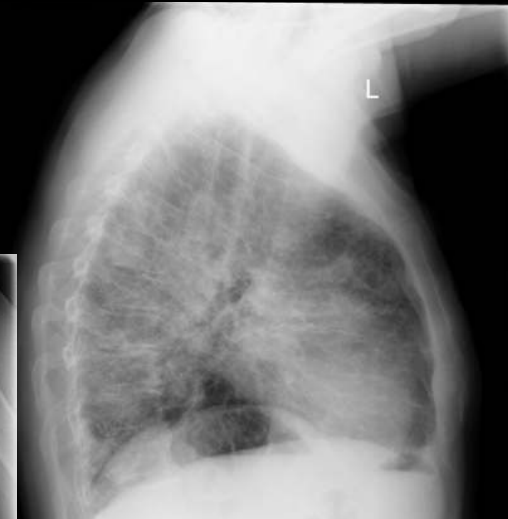
Site	Acute Toxicities	ED Clues / Key Management	Longterm toxicity
Brain / Head & Neck	Radiation necrosis, cerebral edema, mucositis, xerostomia, dysphagia	↑ICP (headache, nausea, focal deficits); treat with steroids (e.g., dexamethasone); airway vigilance if neck edema	Xerostomia Dysphagia Hypothyroid (neck radiation) Cognitive decline
Breast / Chest Wall	Skin changes, rib fractures, lymphedema Radiation Pneumonitis	Painful erythema → suspect radiation dermatitis vs. cellulitis; supportive care, topical steroids if mild	Cardiac or lung toxicity (eg Aortic stenosis, CAD – think Hodgkin’s survivors who had mediastinal XRT)
Abdomen / Pelvis	Enteritis, proctitis, cystitis	Diarrhea, tenesmus, hematuria; hydrate, rule out infection, avoid antidiarrheals if concern for obstruction	Chronic diarrhea, strictures, fistulae Radiation enteritis/proctitis
Spine / Extremities	Myelopathy, fibrosis, secondary malignancy	Progressive weakness or paresthesia → urgent MRI, neurosurg consult	Lymphedema

Red-Flag Presentations

- Acute dyspnea after thoracic radiation → **radiation pneumonitis** (exclude infection).
- Severe dysphagia after neck radiation → **airway edema**.
- Abdominal pain + peritoneal signs after pelvic radiation → **bowel perforation / radiation enteritis**.
- New neurologic deficit after cranial/spinal radiation → **radiation necrosis or edema**

A note on Radiation Pneumonitis

Timing	Delayed. 4-12 weeks (1-3 months post radiation). Way after immunosuppression with chemo
Location	Radiation field
Mimickers	1. Pneumonia (Ground glass opacities) 2. ILD (Interstitial changes) 3. Immunotherapy Pneumonitis (Diffuse + bilateral)
Treatment	Steroids (Prednisone) (Also for immunotherapy pneumonitis)
Bottom Line	1-3 months post radiation with unilateral pneumonia where radiation was done previously? • Written: Steroids • Oral: Empiric CAP abx + Steroids



Source: Dr McKenna – used with permission of patient (who did very well on steroids!)

Q4 2026

A 45-year-old man with metastatic renal cell carcinoma presents with 2 days of polyuria, polydipsia, nausea, and abdominal pain. He has no prior history of diabetes. He is currently receiving pembrolizumab (last dose 3 weeks ago). On exam: HR 118 bpm, BP 102/66 mmHg, RR 26 with deep respirations, dry mucous membranes. Glucose: 25 mmol/L Na 130 mmol/L, K 4.9 mmol/L, HCO₃ 12 mmol/L, Anion gap: 22, Urinalysis pending.

- a) Steroid-induced hyperglycemia; start basal-bolus insulin and fluids
- b) Type 2 diabetes mellitus; start metformin and arrange endocrinology follow-up
- c) **Diabetic ketoacidosis secondary to immune checkpoint inhibitor-induced autoimmune diabetes; admit for IV insulin, fluids, and urinalysis**
- d) Hyperosmolar hyperglycemic state due to infection; give antibiotics and fluids

C is correct. Classic presentation of DKA: hyperglycemia, anion gap metabolic acidosis, Kussmaul respirations. Context of immune checkpoint inhibitor (ICI) therapy, new-onset autoimmune type 1 diabetes and ICI-related toxicity. Steroid-induced hyperglycemia (A) would not cause ketosis/acidosis. T2DM (B) would not present with acute ketoacidosis in a previously healthy adult. HHS (D) features profound hyperosmolality without ketoacidosis.

Palliative Care

- Early Involvement
- Performance Status
- Pain Management + Opioid Rotation
- Symptom Management
- MAID



Q5. Pain Management

Mr. T Pain, a 68yM with metastatic lung CA presents with an acute pathological T8 fracture. He rates his pain as 10/10. He is nauseous and vomiting. AKI with Cr 220. He was taking M-Eslon 120 mg po BID for pain plus 30 mg morphine IR for breakthrough, 7 doses per day before the fracture occurred. In addition to dexamethasone, what treatment will you initiate for his pain control?

- a) Morphine IV 30mg q4h prn
- b) Fentanyl patch 40 mcg q 72h + hydromorphone 2mg sc q1h PRN
- c) Hydromorphone Contin 42mg po BID + 5 mg IR q1h prn
- d) Hydromorphone 8mg sc q4h + 2 mg sc q1h prn
- e) Hydromorphone 6mg sc q4h + 4 mg sc q1h prn



Involving Palliative Care

Palliative-Intent Treatment	Palliative Care*
Oncology Term: (Think Diabetes) Treatment that will control the cancer but can not cure it	Symptomatic management that often coincides with oncological care.
Stopped when patient too frail	Used until the end of life
Bottom Line: Don't delay or avoid palliative care for a patient with metastatic cancer just because they are pursuing disease-directed treatment.** 	

 ** **Benefits include:** better mood, quality of life. Less aggressive interventions near end of life.
Improved survival (**However... Big difference in survival vs Med Onc:** Early Palliative Care (3 months) vs Med Onc (1.5 years to years)¹³. **(NOE)**



Performance Status

Palliative Performance Scale

- 100% (perfectly well) – 0% (death)
- Measure of physical status in palliative care
- PPS 50% (sit/lie throughout day) = only 10% of pts expected to survive >6mo

**Avoid chemotherapy in those
who are unlikely to benefit
(patients ECOG 3-4) ⌘**

ECOG score

Score	Description
0	Asymptomatic
1	Symptomatic Completely ambulatory Able to do light house work
2	<u><50%</u> day spent in bed/chair Capable of all self care Not capable of work
3	<u>>50%</u> day spent in bed/chair Only limited self-care
4	Bedbound
5	Dead

Pain Control

Options

- **Non-opioids** – Acetaminophen, NSAIDs
- **Opioids** – Morphine, Hydromorphone, Fentanyl, etc.
- **Adjuvants** – Pregabalin, Gabapentin, TCAs



Opioid titration

Goal to address most of the baseline pain with long-acting formulations

1. Start with immediate release (IR) formulation Q1-Q4H PRN (q4h PRN for opioid naïve)
2. Low dose → increase to effect/side effects
3. Allow for steady state (~24hrs = 5 half lives)
4. Once steady state achieved, calculate total use in past 24 h
5. Divide 24hr dose into distributed doses of long acting formula
6. Order breakthroughs (= 10% of total daily dose in IR tabs)

Cannabinoids

Nabilone > placebo, but benefits compared to other analgesics not proven (CADTH 2011)



Pain Control

Picking an opioid

- **No comorbidities** – No one right answer. Usually based on physician comfort
- **Renal dysfunction** – Hydromorphone, Methadone, Fentanyl
 - Do NOT use: Morphine, Codeine, Tramadol, Demerol
- **Hepatic dysfunction** – Hydromorphone, Morphine, Fentanyl
 - Do NOT use: Codeine, Methadone
- **Excess opioid-induced itching/urticaria** – Fentanyl

"No Morphine with No Urine"

"No Meth with Failing Heps"

Opioid side effects

- **Common: Constipation, Nausea, Vomiting, Sedation**
- **Rare: Confusion, hallucination, Myoclonus** ← ****Important, lesser-known side effect**
- **Methadone specific: QT prolongation, Drug-drug interactions (CYP3A4 metabolism)**
- If treatment addresses source of pain, need to decrease opioids to prevent toxicity



Opioid Rotation

1. Calculate total daily dose (TDD) of drug
2. Convert TDD to ORAL MORPHINE EQUIVALENT using **equianalgesic table**¹⁹ below
3. Convert TDD to drug of choice
4. Calculate regular/interval dosing – long-acting (BID) or short-acting (q4h)
5. Calculate PRN dose (q1h) based on 10% of TDD or 50% of q4h dose

	Equivalence to 30 mg oral morphine	To convert to PO morphine, multiply by	PO : IV/SC	Effect Duration
Morphine	30 mg	1	2:1	2-4 hrs
Oxycodone	20 mg	1.5	N/A	3-4 hrs
Hydromorphone	6 mg	5	2:1	2-4 hrs
Transdermal Fentanyl	60-90 mg morphine = 25 mcg/hr 120-180 mg morphine = 50 mcg/hr			48-72 hrs

Clinical Pearl: If switching meds due to toxicity (i.e. myoclonus), consider 25-50% dose reduction to allow for cross tolerance → 50% for well controlled pain, 25% for uncontrolled pain.

Symptom Management

Dyspnea

- Stepwise approach: **Non-pharmacologic therapies** (i.e. fan, open window, walker) → **oxygen** (if hypoxic) → **Opioids**²⁰
- Oxygen ineffective for non-hypoxemic dyspnea[⌘]

Nausea and Vomiting

- Direct treatment towards etiology
- Unknown etiology – Dopamine antagonist favored (haloperidol, olanzapine) or agents used for chemotherapy-induced nausea and vomiting

Delirium

- Risperidone/haloperidol do not alleviate distress at end of life and tend towards harm (JAMA 2017)
 - May slightly worsen delirium symptoms (low quality evidence, Cochrane Review 2020)
 - May increase adverse side effects (EPS symptoms) (low-mod quality)

Cachexia:

- Refer for counseling + assessment re: high protein, nutrient dense food, advise **against** extreme diets/fad diets
- Do **NOT** offer enteral tube feeding or parenteral nutrition to manage cancer cachexia
- **Olanzapine now recommended at low dose.**
- *ponsegromab (GDF15 monoclonal antibody) may be coming to Canada soon however not currently in guidelines.

Medical Assistance In Dying – Legal since 2017

Criteria- Need to meet ALL FIVE

Personal Attributes	1. Eligible for health services funded by the province/ country 2. 18+ years old and is capable* 3. Make a voluntary request for MAID (free from outside pressure or influence) 4. Provide informed consent for MAID
Disease Attribute	5. Grievous and Irremediable Physical, NOT mental illness**~ 1. Serious and Incurable illness 2. Advanced decline that CAN'T be reversed 3. Suffering but ways of improvement are not acceptable to patient



* Can't be in state of delirium/ dementia/ encephalopathic (ex. From organ failure from cancer)

** Mental Health as sole underlying condition still excluded- Eligibility date moved to 2027.

~ It used to be "death is reasonably foreseeable". This was repealed in 2021. (ex. ALS patients would previously be excluded but now are not)

NEW MAID LAW
CAME INTO FORCE ON MARCH 17, 2021

**Stream 1: Death is reasonably
Foreseeable ***

1. Request in writing, signed by independent **witness**
- **2 independent MD/NPs** must confirm all eligibility criteria met
- Person must **be given opportunity to withdraw consent**, and **confirm consent expressly before receiving MAID**



*MAID can happen as quickly as 1-2 days depending on logistics. 10 day reflection was removed

REQUEST FOR MAID



ELIGIBILITY CRITERIA

- HAVE A GRIEVOUS AND IRREDEMIABLE MEDICAL CONDITION
- "REASONABLE FORESEEABILITY OF NATURAL DEATH" CRITERION REPEALED
- ELIGIBILITY FOR PERSONS SUFFERING SOLELY FROM MENTAL ILLNESS IS TEMPORARILY EXCLUDED UNTIL MARCH 17, 2023 (EXPERT REVIEW WILL MAKE RECOMMENDATIONS ON PROTOCOLS, GUIDANCE AND SAFEGUARDS FOR MAID FOR PERSONS SUFFERING FROM MENTAL ILLNESS BY MARCH 17, 2022)



**EXISTING & EASED
SAFEGUARDS**
FOR ELIGIBLE PERSONS
WHOSE DEATH
IS REASONABLY
FORESEEABLE



**NEW & STRENGTHENED
SAFEGUARDS**
FOR ELIGIBLE PERSONS
WHOSE DEATH
IS **NOT** REASONABLY
FORESEEABLE

**FINAL CONSENT OR
POSSIBLE WAIVER
OF FINAL CONSENT**
FOR ELIGIBLE PERSONS
UNDER CERTAIN
CIRCUMSTANCES

**FINAL CONSENT
TO BE PROVIDED BEFORE
ADMINISTRATION OF MAID**

MAID

Stream 2: Death is NOT foreseeable

- Request in writing, signed by independent **witness**
- **2 independent MD/NPs** must confirm all eligibility criteria met **AND consult with expert in the medical condition** (if MD/NP is unfamiliar with it)
- The person must be informed of means to alleviate suffering and be offered consults with experts in this (incl. counselling, pall care...)
- **The eligibility assessments must take at least 90d**, but this period can be shortened if pt about to lose capacity
- Person **must be given opportunity to withdraw consent**, and confirm consent expressly before receiving MAID

MAID: Final Consent Requirement

Pt does not have to provide consent immediately before provision of MAID if :

1. Pt has been assessed and approved for MAID
2. Pt was at risk of losing decision making capability prior to receiving MAID and was made aware of that risk
3. Person makes arrangement in writing with their practitioner to waive final consent, and according to which **their practitioner will provide MAID on their preferred date if they have lost capacity** (*“Audrey’s amendment”*)

MAID legislation... changes pending

Advanced Request Legislation = Not Legal Yet

Scenario: A physician has early dementia and requests, when capable, that when she is dependent for an ADL like toileting or cannot recognize her family she would like to have MAID.

Under current Canadian Criminal Code legislation, this is not allowed as patient would not be capable at time of MAID and cannot give an 'advanced date' as time from decision to incapacity could be months or years in this scenario, depending on cause of dementia.

- **Exception: in Quebec - as of October 30, 2024 with Bill 11 amending Act Respecting End-of-Life care.** Quebec physicians can provide MAID in Quebec (only Quebec) under an advanced request for those diagnosed with a serious and incurable illness leading to incapacity (e.g., Alzheimer's).
- This means there is a discrepancy between Quebec and federal MAID laws
 - However Health Minister (federal) has indicated they will not challenge Quebec in their legislation
- This is an area of ongoing debate and legislation so is an unlikely scenario on the Royal College in spring 2026

MCQ 5. Pain Management

Mr. T Pain, a 68yM with metastatic lung CA presents with an acute pathological T8 fracture. He rates his pain as 10/10. He is nauseous and vomiting. AKI with Cr 220. He was taking M-Eslon 120 mg po BID for pain plus 30 mg morphine IR for breakthrough, 7 doses per day before the fracture occurred.

What will you initiate?

- a) Morphine
- b) Fentanyl patch
- c) Hydromorphone
- d) Hydromorphone
- e) **Hydromorphone**

Opioid Rotation Calculation

- 1) TDD = 450 mg morphine po
- 2) Divide by 2 to get sc/IV equiv: 225 mg
- 3) Due to renal toxicity, convert to non-morphine (eg) hydromorphone – divide by 5 = 45 mg total daily iv/sc dose
- 4) When converting dose reduce to avoid toxicity w AKI = 6mg q4 (36 mg per day)
- 5) Calculate PRN = 10% TDD

Answer E: Hydromorphone 6mg sc q4h + 3mg sc q1h prn

Wrong Answers: Morphine shouldn't be used in AKI. Patient is vomiting so po hydromorphone not good. Fentanyl patch takes 12h to 'kick in' so not ideal in this setting of severe acute pain

BONUS MCQ (2023): Pain Management

A 56yF with metastatic ovarian CA presents with a malignant bowel obstruction. She has been vomiting for 3 days and unable to keep anything down. She has an acute kidney injury, Cr 217. She rates her pain as 10/10. She was taking M-Eslon 60 mg po BID for pain plus 10mg morphine IR for breakthrough, taking 4 doses per day before this bowel obstruction occurred. In addition to IV hydration and anti-emetics, what treatment will you initiate for her pain control?

- a) Morphine IV 10mg q4h prn
- b) Hydromorphone CR 6mg po BID + 2mg IR q2h prn
- c) Hydromorphone 2mg sc q4h + 1mg sc q1h prn
- d) Hydromorphone 4mg sc q4h + 1.5 mg sc q1h PRN
- e) Fentanyl patch 37.5 mcg q 72h + hydromorphone 2mg sc q1h PRN

Opioid Rotation Calculation - Answer C

1) TDD = 160mg morphine po

2) Divide by 2 to get sc/IV equiv: 80mg

3) Due to renal toxicity, convert to non-morphine (eg) hydromorphone – divide by 5 = 16mg total daily iv/sc dose
4) When converting dose reduce to avoid toxicity w AKI = 2mg q4 (12mg per day)

5) Calculate PRN = 10% TDD

Questions?

Many practice MCQs available in your BONUS materials for self-study and on FLEXIQUIZ online



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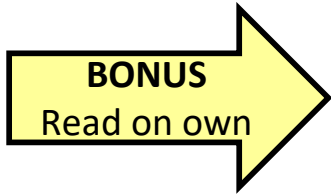
BONUS

Read on own

Self study: Metastases Trivia

Lymph Node Metastases

- **Right supraclavicular** nodes
 - **Lung**, Esophageal
- **Left supraclavicular** nodes ("Virchow's node")
 - **Abdominal malignancies** (via the thoracic duct) – Gastric, gallbladder, **pancreas**, kidneys, testicles, ovaries, prostate
 - Ipsilateral breast & lung
- **Umbilical** nodes ("Sister Mary Joseph Node") = Intra-abdominal/pelvic metastases
 - GI – **Gastric**, colon, pancreas
 - Gynecologic – **Ovarian, endometrial**
 - Unknown primary



Self study: Metastases Trivia

Bone metastases

- Osteoblastic (ie. Sclerotic)
 - **Prostate**, SCLC, Carcinoid, Hodgkin lymphoma
- Osteolytic
 - **Multiple myeloma**, NSCLC, RCC, Melanoma, Thyroid, Non-Hodgkin lymphoma
- Mixed
 - **Breast**, GI, Squamous cell carcinomas (NSCLC, H&N, Cervical ca)

BONUS

Read on own

Self-Study: Cancer of Unknown Primary

!!

Histology	Clinical Features	Work-Up	Suggested treatment
"Adenocarcinoma"	Woman, Axillary adenopathy	Mammo, MRI breast	Treat as breast ca
	Woman, Peritoneal carcinomatosis	CA-125, TVUS	Treat as ovarian ca
	Man, Elevated PSA, Bone metastases	PSA, Bone scan	Treat as prostate ca
"Squamous cell carcinoma"	Cervical adenopathy	Pan-endoscopy, PET	Treat as H&N ca
	Man, Inguinal adenopathy	Lower endoscopy	Treat as anal cancer
	Woman, Inguinal adenopathy	Pap smear, TVUS, endoscopy	Treat as cervical cancer <i>or</i> anal cancer
"Poorly-differentiated carcinoma"	Young man, Midline tumour, Elevated hCG/AFP	Scrotal ultrasound	Treat as testicular primary

BONUS
Read on own

Self Study: Renal Cell Carcinoma



RCC once called “The Internist’s Tumour” due to the variety of paraneoplastic phenomena. Remember to consider as a differential if you see any constellation of the following:

- Hypercalcemia
- Hypertension
- Polycythemia
- Cushing’s
- Galactorrhea
- Amyloidosis
- Coagulopathy/ Thrombosis
- Stauffer’s Syndrome

Classic Triad (occurs in less than 15% of patients)

- palpable mass
- Hematuria
- Flank pain

BONUS
Read on own

Chemo Side Effects: Diarrhea and Mucositis

Side Effects	TOP Differential	Clinical Presentation	Management
Diarrhea*	1. Systemic Therapy (Chemo) 2. Immunotherapy 3. Infection (C.diff) 4. Overflow diarrhea	• Watery diarrhea • Can be bloody • Abdo pain (Enteritis) • Often large volume (except for number 4)	1. Rule out C.Diff... then 2. Loperamide** 3. Lomotil 4. Octreotide
Mucositis	1. Thrush 2. HSV mouth sores	• Oral ulcers • burning tongue/ mouth • Odynophagia	1. Baking soda rinses 2. Viscous lidocaine 3. Magic Mouthwash (nystatin + lidocaine)

***Sometimes Chemotherapy Diarrhea balances out the hydromorphone constipation which is nice...**

****Aggressive Loperamide regimen:**

- 2 tabs (4mg) with 1st loose stool, 1 tab with EVERY loose stool until no more x 12 hours.
- Max 8 tabs / 24 hours.

BONUS
Read on own

Infamous Chemo Side Effects

Drug	Management	Implicated Chemo Drugs
Alopecia	No great therapy. Small risk of permanent hair loss Cold Capping (Preventative)- Mixed results	-rubicins (Breast, Lymphoma) -tecans (Lung, Gyne) -taxels (Breast, Lung, Gyne)
Rash	1. Body: Betaderm. Face: Hydrocort. Widespread: Systemic steroid 2. Moisturizer 3. Sunscreen/ Sun protection/ Sun avoidance	- 5FU / Cape (GI, Breast)
Neuropathy (Glove and Stocking)	No great therapy. Risk of permanent neuropathy.	-taxels (Breast, Lung, Gyne, Esophagus) -platins (Lung, Gyne)
Myelosuppression	Feb Neut Platelet issues rare. Transfuse Hb PRN.	Any chemotherapy
Secondary Malignancies (Generally haem)	No great preventative options aside from changing chemotherapy	-fosfamide (Breast, Lymphoma) - rubicins (Breast, Lymphoma)

Bonus 1 2026

A 64-year-old woman with metastatic melanoma on nivolumab presents to the ED with vague chest discomfort and mild dyspnea. She received her third cycle 10 days ago. On exam she has a HR 108 bpm, BP 100/64 mmHg, SpO₂ 95% RA, JVP not elevated, no peripheral edema. Investigations show a ECG: sinus tachycardia with new incomplete right bundle branch block. High-sensitivity troponin I: 250 ng/L (previously <5 ng/L), BNP: 450 pg/mL. Her Chest X-ray was clear. Echo completed in ED showing LVEF 50% (baseline 65%), no pericardial effusion

- A. Admit to oncology, start high-dose steroids immediately, and arrange cardiology review in the morning
- B. Discharge with close outpatient follow-up since symptoms are mild and LVEF is preserved
- C. Urgently involve cardiology overnight for suspected immuno-oncology-related myocarditis, initiate high-dose steroids, and co-manage inpatient
- D. Observe on telemetry overnight without steroids and repeat troponin/echo in the morning

Bonus 2 2026

Miss Liv Inwell is a 45F from Cote d'Ivoire who presents for routine cancer screening. She reports a recent diagnosis of HBV (Sag+), family history of colon cancer in her father at age 70, She smoked 1 ppd from age 18 onward. She loves to tan. No family history of breast cancer. What cancer screening do you recommend?

- a) Mammogram, FIT testing, Abdo U/S, AFP, annual skin exam.
- b) FIT testing, AFP, low dose CT Thorax, PSA
- c) Mammogram, Abdo U/S, AFP, colonoscopy, low dose CT Thorax
- d) Mammogram, FIT testing, AFP, low dose CT Thorax
- e) Colonoscopy, Abdo U/S, AFP

BONUS MCQ 3

Mrs. Celine Dionne is a 30F on pembrolizumab for colon cancer x 1 year presents to office with progressive fatigue, dyspnea, feeling cold and clammy. You note bilateral pitting edema 3+. HR is 115, BP is 100/50, SPO2 90%RA, Afebrile. TSH 20. CXR shows non-specific interstitial changes. What would be the best NEXT investigation?

- A. Echocardiogram
- B. Free T3 and T4
- C. Blood gas
- D. Blood culture x 2

BONUS Q4

A 61M with a 40-pack-year smoking history presents with progressive fatigue, proximal leg weakness, and occasional dizziness when standing. He reports dry mouth and constipation. Symptoms have been present for several months. On exam: BP 96/64 mmHg sitting, 78/54 mmHg standing HR 82 bpm. Neuro exam: symmetric proximal lower extremity weakness; strength improves with repeated testing Reflexes: reduced but improve after brief exercise. No cushingoid features, no edema. Labs: Na: 137 mmol/L, K: 3.8 mmol/L Random cortisol: 350 nmol/L Chest CT: large hilar mass with bulky mediastinal nodes.

Which of the following paraneoplastic syndromes most likely explains his presentation?

- A. Syndrome of inappropriate antidiuretic hormone secretion (SIADH)
- B. Lambert–Eaton myasthenic syndrome (LEMS)
- C. Ectopic ACTH secretion (Cushing’s syndrome)
- D. Hypercalcemia of malignancy (PTHrP)

Bonus Qs 2025-6 Answers

- 1) C- Immuno-oncology-related myocarditis is rare but potentially fatal, with high risk of arrhythmia and cardiogenic shock. Presentation can be subtle (mild troponin rise, preserved/mildly reduced EF, nonspecific ECG). Steroids alone are not enough → early cardiology involvement is critical, even overnight. **Answer C**
- 2) **E- Colonoscopy, Abdo U/S, AFP.** Wrong answers: A, B, C – Colonoscopy for high risk. She has no cervix. Melanoma screening not routinely recommended. D – No need for CT given young age
- 3) **A – ECHOCARDIOGRAM.** **Wrong answers:** B – While technically correct, wouldn't be the **BEST** test. Myxedema generally would be with much higher TSH C – While this would provide you with oxygenation information, wouldn't contribute much to the diagnosis. D – There's no evidence sepsis is the most likely clinical picture here. Correct answer: A – Safety exam- Rule out most dangerous thing which would be immunotherapy-related myocarditis
- 4) **Answer B LEMs.** LEMS is a paraneoplastic autoimmune disorder (antibodies against presynaptic voltage-gated calcium channels) strongly associated with SCLC. Key features: Proximal weakness that improves with repeated effort, Autonomic symptoms: dry mouth, constipation, orthostatic hypotension, Hyporeflexia that improves after exercise. Note, Hypercalcemia (D) is squamous cell carcinoma, not SCLC.

BONUS Q1 2025

40F nonsmoker presents with stage IV lung cancer, she has shaved her hair in preparation for chemotherapy. You are getting her cancer worked up so she can go see her oncologist. Which of the following statement is correct?

- A. She will need molecular testing as she may not need chemotherapy as treatment.
- B. The presence of pleural effusion means there's definite metastatic lung cancer.
- C. Palliative care referral is only required near the last 3 months of life
- D. Stage IV lung cancer is one of the few cancers that are curable at stage IV setting.

BONUS Q2 2025

50M, obese, with history of hypertension, dyslipidemia, presents with right axillary lymph node enlargement, mixed sclerotic and lytic bone metastases. Biopsy of the lymph node reveals adenocarcinoma. What is the best NEXT investigation?

- A. PSA and DRE
- B. Mammogram
- C. Colonoscopy
- D. PET scan

BONUS Q3 2025

18M with end-stage metastatic colon cancer. He is clearly in distress and his family is begging you to assess him for MAID. You find him at the bedside but he remains delirious, same as he was over the last month. What is the best management?

- A. Consent is obtained from patient and MAID is performed.
- B. Consent is obtained from power of attorney/ substitute decision maker and MAID is performed.
- C. Consent is not possible and thus MAID is not allowed.
- D. Consent is assumed given his situation and thus MAID is performed.

BONUS Q4 2025

74F, ultramarathon runner otherwise healthy, presents to the emergency department with acute onset dyspnea. On exam her HR was 120, BP 110/53, spO2 83% RA, JVP elevated, 2+ peripheral edema. CXR showed bilateral interstitial changes. You note that she had a mastectomy scar over her left chest and was notified she had breast cancer 10 years ago. What's the likely cause of her current symptoms?

- A. Herceptin
- B. Myocardial infarction
- C. Doxorubicin
- D. Pneumonia

BONUS Q5 2025

68F presents to clinic with 1 day of fever, confusion, diplopia, and subsequently seizure that resolved with levetiracetam. On reviewing the chart you see that she has metastatic melanoma and is currently on immunotherapy with Nivolumab. MRI head is performed which showed no intracranial abnormalities. Rest of vitals all normal. CBC, lytes, extended lytes, Cr, LFTs all normal. What is the best next step?

- A. LP to rule out meningitis / encephalitis
- B. Order serum anti-Hu antibody
- C. Start pulse steroids
- D. EEG

BONUS 2025 MCQ ANSWERS

1. A- Lung cancer management has evolved and thus some mutations like EGFR can be targeted by oral pills like Osimertinib- significantly prolonging their survival. Pleural effusion needs to prove malignant. Early palliative care is critical. Stage 4 lung cancer is NOT curable.
2. B- Male can have breast cancer too. This history suggests ++ estrogen due to high adiposity. Prostate shouldn't have lytic bone metastases. Colon CA doesn't generally give axillary LN. PET scan not indicated upfront when there can be more specific investigation.
3. C- Consent for MAID HAS TO be from patient, POA and SDM not allowed.
4. C- She likely had chemotherapy for her breast cancer, which can cause delayed and permanent cardiotoxicity. Herceptin heart failure occurs on treatment, doesn't lead to problems years down the line. Myocardial infarction on ddx but there's a much more compelling answer and she is an ultramarathon runner with no significant cardiac risk.
5. A- She likely has immune-related encephalitis. LP is needed. Anti-Hu is part of SCLC paraneoplastic antibody. No dx yet so no steroids. EEG would not lead to the diagnosis.

2024 LECTURE MCQ1. Screening

Miss Liv Well is a 42F presents for routine cancer screening. She reports a recent diagnosis of MASH (metabolic dysfunction associated steatohepatitis) Childs Pugh A cirrhosis, family history of colon cancer in her father at age 70, She smoked 1 ppd from age 18 onward. Her cervical cancer screening is up to date. What cancer screening do you recommend?

- a) FIT testing, AFP, low dose CT Thorax,
- b) Colonoscopy, Abdo U/S, AFP
- c) Mammogram, Abdo U/S, AFP, colonoscopy, low dose CT Thorax
- d) Mammogram, FIT testing, AFP, low dose CT Thorax
- e) Mammogram, FIT testing, Abdo U/S, AFP, low dose CT Thorax

2024 LECTURE MCQ 2

Mr. Remy Kade is a 35M with history of uncontrolled Crohn's disease is admitted to your service with RLQ abdominal pain. CT C/A/P shows small lesions throughout his lungs and liver, suspicious for metastases. Hb 100, Plt 200, WBC 9. CRP is normal. What is the best next step?

- A. Colonoscopy, CEA
- B. Prednisone 1mg/kg daily
- C. Abdo U/S for liver biopsy, AFP
- D. PET scan, CA 19-9, CEA, CA 125
- E. DRE, PSA
- F. Call Medical Oncology

2024 LECTURE MCQ 3

Ms. Sega Rett is a 55F with LUL lung cancer, received concurrent chemoradiation. 7 weeks post treatment presents with 2 weeks of non-productive cough and low grade fever. Appears well on exam, satting 94% on room air but desat to 88% with exertion, no wheezing on exam, crackles LUL. Chest x-ray showed LUL consolidation. CT thorax shows LUL lung mass with surrounding ground glass. There's no evidence of pleural effusion or lymphadenopathy. What is the treatment of choice?

- A. Ceftriaxone + Azithromycin
- B. Azithromycin
- C. Apixaban
- D. Fluticasone/Salmeterol
- E. Prednisone



2024 LECTURE MCQ 4. Pain Management

Mr. T Pain, a 48yM with metastatic lung CA presents with an acute pathological T8 fracture. He rates his pain as 10/10. He is nauseous and vomiting. AKI with Cr 200. He was taking M-Eslon 90 mg po BID for pain plus 20 mg morphine IR for breakthrough, 3 doses per day before the fracture occurred. In addition to dexamethasone, what treatment will you initiate for his pain control?

- a) Morphine IV 20mg q4h prn
- b) Hydromorphone Contin 12mg po BID + 5 mg IR q1h prn
- c) Hydromorphone 3mg sc q4h + 2 mg sc q1h prn
- d) Hydromorphone 5mg sc q4h + 3 mg sc q1h prn
- e) Fentanyl patch 50 mcg q 72h + hydromorphone 2mg sc q1h PRN

BONUS Q5 2024

50F with Diffuse Large B Cell Lymphoma, recently received prednisone 4 days. She comes into ER feeling unwell. HR is 132, BP 90/68, Temp 38.5, RR 20. Bloodwork: Neutrophils 5, K 6.3, Ca 2.1, Cr 180, PO4 2.2, Uric acid 380, LFT normal. CT shows shrinking lymph nodes compared to the scan a month ago. She has no other significant past medical history and takes no medications. What should you do next?

- A. Start IV fluids and send off G6PD testing
- B. Start IV fluids and start Allopurinol
- C. Start IV fluids and start Rasburicase
- D. Start IV fluids and Piptazo

BONUS Q6 2024

A 55M with recently identified large RUL lung mass, with trace right pleural effusion presents to your clinic for further workup. He is currently asymptomatic. What is the next best step?

- A. Thoracic surgery for thoracentesis.
- B. Consult early palliative care as early referral has survival advantage.
- C. Urgent referral to medical oncology to get opinion on treatment.
- D. IR for lung mass biopsy.

BONUS Q7 2024

A 34 premenopausal female with breast cancer on tamoxifen recently presented to clinic for routine check-up. She is having intermittent vaginal bleeding that is not part of her period. She wants to switch over to Letrozole. What would you advise her, assuming her medical oncologist is not available?

- A. Switch to letrozole.
- B. Stop the tamoxifen. Organize an endometrial biopsy
- C. Reassure her that tamoxifen confers cardiovascular benefit, and to continue the tamoxifen
- D. Stop the tamoxifen. Organize a pap smear

BONUS Q8 2024

A 34F with triple negative breast cancer on chemotherapy. She came into ER with dizziness, vomiting and ataxia. On brain MRI she was found to have a left cerebellar lesion and bilaterally enlarged ventricles with obstructive hydrocephalus. What is the next best step?

- A. Lumbar puncture to relieve the pressure
- B. Dexamethasone
- C. Acetazolamide
- D. Neurosurgery for shunt

BONUS 2024-5 MCQ ANSWERS

1. B- Colonoscopy for high risk. No need for CT given young age
2. A – Most likely colon adenoCA with mets to lung and liver. F – Last resort on oral exam which is still acceptable. Steroids won't help with dx. AFP not indicated- goal is to find the instigating cancer. PET not indicated prior to cancer-specific test. Prostate typically in older men. There's more likely dx.
3. E- Oral steroids are needed to treat radiation pneumonitis so this is the best answer. In real life however, would cover both pneumonia and radiation pneumonitis.
4. C- Morphine shouldn't be used in AKI. Patient is vomiting so po hydromorphone not good. Fentanyl patch takes 12h to 'kick in' so not ideal in this setting of severe acute pain
5. C- She's a female so not likely G6PD. She's unwell, hemodynamic unstable, there's signs of treatment response with biochemical features of TLS. Treat!
6. D- Need bx of primary to identify the malignancy first (Effusion can wait. May be reactive). In addition, thoracic surgery doesn't do paracentesis. No pathology so don't refer to med onc yet. Palliative care not appropriate- we don't even know if this is cancer yet.
7. B- While tamoxifen has theoretical CV benefit, the risk of endometrial cancer is real. Biopsy is the gold standard. Pap smear is for cervical cancer screening. Need to stop the tamoxifen but can't start AI as this now boosts estrogen in pre-menopausal woman!
8. B- Leptomeningeal disease is deadly and often occurs with breast cancer and lung cancer. Shunting doesn't work well as the cancer cells just plug up the shunt. LP will cause herniation. Dexamethasone would be first line to help reduce some swelling and buy time for chemo.



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